COMPUTER VISION TECHNICAL EVALUATION | RESEARCH & ENGINEERING

Egg Instance Segmentation and Multi-Object Video Tracking

Candidate: Aaditya Jain **Evaluation Role:** CV Intern Assessment

Repository: github.com/Aaditya-Jain-01/egg-segmentation-and-tracking

1. Dataset Partitioning & Data Leakage Mitigation

The benchmark dataset comprises 110 macroscopic RGB images capturing eggs situated in diverse real-world environments, featuring prominent variations in hand grasp poses, skin tones, surface reflections, and ambient illumination. The dataset was partitioned into a development set (88 images, 80.0%) and an independent evaluation set (22 images, 20.0%), adhering strictly to the required 4:1 ratio.

Data Leakage Prevention: Scraped visual datasets derived from burst captures or continuous video sequences exhibit severe temporal autocorrelation. Uniform random sampling distributes near-identical consecutive frames across both partitions, causing feature memorization and falsely inflating validation metrics. To prevent leakage, filenames were grouped into sequential block clusters based on scene prefixes and timestamp windows. Splitting was enforced at the cluster boundary, guaranteeing that entire capture sequences remained strictly isolated within either the development split or the evaluation split.

2. Task 1: Pretrained Foundation Model Segmentation

Meta AI's Segment Anything Model (SAM) with the Vision Transformer Base (`ViT-B`, 86M parameters) backbone was utilized to establish high-fidelity pseudo-ground truth masks across all 110 images.

Candidate mask proposals were synthesized via `SamAutomaticMaskGenerator`. To ensure thorough spatial coverage across macroscopic objects while eliminating redundant inference latency on standard compute, an evaluation grid of `12 × 12` (144 evaluation seeds) was utilized with global-scale encoding (`crop_n_layers=0`). Candidate masks were ranked and filtered using a composite geometric fitness score:

> **Geometric Scoring Function:**
> $\text{Score} = 0.5 \times \text{Circularity} + 0.5 \times \text{Solidity} + 0.3 \times \text{Pred_IoU}$
> subject to: $0.02 \times A_{img} \leq A_{mask} \leq 0.85 \times A_{img}$ and $\text{Solidity} \geq 0.85$

This formulation rigorously discards hand backgrounds, fingertip artifacts, and room-scale bounding boxes, isolating the prolate spheroidal egg boundary as binary masks for evaluation benchmarking.

3. Task 2: Classical Image Processing Experiment Log

Developing a purely rule-based classical segmentation pipeline to approximate deep foundation model quality requires balancing foreground shell contrast against hand occlusions and surface glares. The development trials are documented below:

Trial #	Technique & Parameters	Hypothesis / Motivation	Dev Results	Verdict

Trial 1	**Strict Geometric Gate & Ellipse Fit**
	• Otsu on CIELAB `L*` & HSV `V`
	• HSV skin mask subtraction
	• Ellipse intersection via `fitEllipse`
	• Solidity > 0.80, Circularity > 0.45
		Eggs conform closely to prolate spheroids. Forcing boundary intersection with a fitted mathematical ellipse should prune irregular hand/finger extensions.	`IoU: 0.2217`
			`Dice: 0.2479`
			`Prec: 0.2721`
			`Rec: 0.2437`
			`Acc: 0.8913`
				DISCARDED
				Extreme penalty on partial occlusions. When fingers crossed the egg, non-elliptical boundaries caused empty mask fallbacks (IoU = 0.0).

Trial 2	**Multi-Space Decoupled Fusion & Hole-Fill**
	• Candidate = $(L^*_{Otsu} \cup Gray_{Otsu}) \cap V_{Otsu} \cap (S < 90)$
	• Dual morphology: Open (5×5), Close (19×19)
	• Full flood-fill internal closure
	• Center-proximity weighted contour ranking
		Decoupling luminance from saturation isolates low-chroma shells from saturated skin tones. Large morphological closing and flood-filling seals specular reflections.	`IoU: 0.3253`
			`Dice: 0.4183`
			`Prec: 0.3741`
			`Rec: 0.7426`
			`Acc: 0.6716`
				SELECTED
				Tripled recall from 0.24 to 0.74. Recovered complete egg body coverage across lighting variations and eliminated false-negative dropouts.

Trial 3	**Bilateral Edge Filter & Skin Gating**
	• Bilateral Filter (`d=9`, σ_r=75, σ_s=75)
	• YCrCb skin rejection (`135 < Cr < 175`)
	• Saturation floor (`35 < S < 190`)
	• Morphological close (15×15)
		Bilateral filtering smooths skin creases while preserving sharp shell edges; targeted `Cr` gating prunes hand boundary bleed.	`IoU: 0.2266`
			`Dice: 0.2561`
			`Prec: 0.2589`
			`Rec: 0.3117`
			`Acc: 0.8657`
				DISCARDED
				Rigid chrominance bounds aggressively excised warm-toned brown shells and shaded shell bottoms, causing recall to drop sharply.

4. Task 3: Quantitative & Qualitative Comparative Analysis

Treating Task 1 SAM masks as ground truth and the optimized classical algorithm (Trial 2) as predictions, comprehensive evaluation metrics were computed across both dataset partitions:

Dataset Split	Mean IoU (Jaccard)	Dice Coeff (F1)	Precision	Recall	Pixel Accuracy
Development Split (88 images)	**0.3253**	**0.4183**	0.3741	0.7426	0.6716
Evaluation Split (22 images)	**0.2761**	**0.3737**	0.3424	0.8163	0.5530

Qualitative Breakdown & Failure Mode Diagnostics

Comparative inspection of generated error heatmaps (True Positives: Yellow, False Positives: Red, False Negatives: Green) reveals three distinct physical failure mechanisms inherent to classical rule-based segmentation:

1. **Low-Contrast Hand Contact Bleed (False Positives):** In images where subjects cradle light-brown or unwashed eggs, the palm's inner skin surface shares near-identical luminance and chrominance profiles with the shell. Classical intensity thresholding merges the egg boundary into the palm, resulting in false-positive spillover along the lower contour.
2. **Specular Highlight Fragmentation:** Direct overhead lighting induces severe localized specular reflections. These saturated white glares create high-gradient inner boundaries that can sever binary masks into annular rings unless bridged by extensive morphological closing.
3. **Severe Digit Occlusion (False Negatives):** When multiple fingers wrap across the shell's ventral axis, classical contour detection splits the foreground into several disconnected components. Area-ranking heuristics retain only the single largest fragment, discarding the remaining segments.

5. Task 4: Multi-Object Video Tracking Implementation

The objective of Task 4 is the autonomous detection and continuous identity-preserving tracking of individual eggs throughout the evaluation video sequence.

Tracking Pipeline Architecture:

- **Per-Frame Detection:** Each incoming frame is transformed into CIELAB color space. Adaptive Gaussian thresholding on the `L*` channel isolates candidate foreground regions, which are geometrically validated using solidity (`> 0.80`) and circularity (`> 0.40`) gates.
- **Centroid Distance Association:** An assignment cost matrix is constructed computing pairwise Euclidean distances between established tracklet centroids and newly detected candidate centers. Optimal matches within a spatial gating threshold ($d_{max} = 80$ px) update object positions.
- **Tracklet Persistence:** To handle temporary occlusions and transient frame blurs, deregistration is buffered across a grace period of 20 consecutive missed frames.
- **Dynamic Visual Output:** Each unique egg is assigned a persistent identifier (`Egg #1` , `Egg #2`), a distinctive color-coded bounding box, and a smoothed 30-frame historical trajectory trail.

> **Public Video Output Link:**
> Access the full processed tracking video output (`tracked_output.mp4`) on Google Drive:
> **https://drive.google.com/file/d/1_r4a-0JjMyHaAlcRM1OdY_KvlZPC4YFP/view?usp=sharing**

6. AI Tools Disclosure & Reproduction Guide

Disclosure: In accordance with assessment instructions, AI coding assistants were utilized as productivity tools for environment troubleshooting (resolving headless Matplotlib rendering backends on Windows) and syntax drafting. All analytical reasoning, mathematical scoring criteria, pipeline architecture, and experiment log design were developed and validated independently.

Reproduction Commands:

```
git clone https://github.com/Aaditya-Jain-01/egg-segmentation-and-tracking.git
cd egg-segmentation-and-tracking
python -m venv venv && venv\Scripts\activate
pip install -r requirements.txt
curl.exe -L -o weights/sam_vit_b_01ec64.pth
https://dl.fbaipublicfiles.com/segment_anything/sam_vit_b_01ec64.pth
python 01_split_data.py && python 02_task1_sam_segmentation.py && python
03_task2_classical_segmentation.py && python 04_task3_evaluation.py && python
05_task4_video_tracking.py
```